Roomify

Deployment

Proposed hosting, cost and steps (proposal only, not deployed)

Team Bootcamp2Roomify
Release v1.0.0 (demo mode)
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github.com/Bootcamp2Roomify/roomify_frontend · roomify_backend · roomify_cv

Deployment

Status (2026-10-26): approved, configs committed, Neon database created, not yet deployed. Blocked on (1) a payment method on the Fly.io account (`fly apps create` answers "We need your payment information") and (2) a bucket-scoped AWS key. The judging build runs with `AI_DEMO_MODE=true` (see step 3).

Recommendation: Vercel + Fly.io + Neon + existing S3

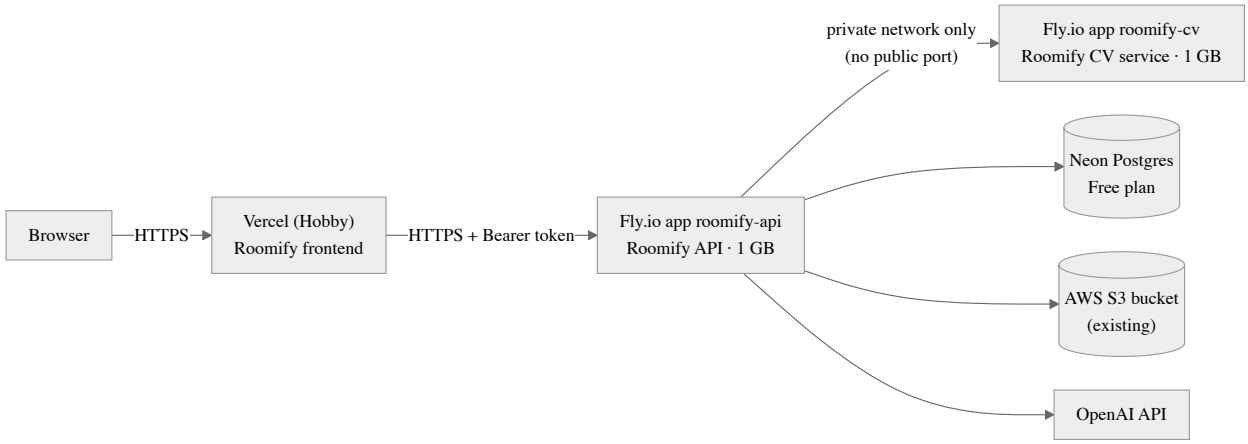


Figure 1. Proposed Roomify deployment: Vercel for the frontend, Fly.io for the API and a private CV service, Neon Postgres, the existing S3 bucket and the OpenAI API.

Why this shape:

- **Each service already ships as a container** (backend and CV Dockerfiles are pinned and tested), so the hosts only run images: no build-system work.
- **The CV service gets no public address.** It has no authentication and an `image_url` input; Fly's private network lets only the backend reach it.
- **Long AI calls fit.** Image edits can take up to ~2 minutes. Vercel Hobby functions with Fluid compute allow 300 s; the backend's AI timeout is 180 s. The browser never calls the backend directly (the Next bridge does), so no CORS changes are needed.
- **Measured memory fits 1 GB machines:** CV container 374 MiB after inference, backend 424 MiB after start-up (Docker, 2026-10-24).

Cost (list prices checked 2026-10-24)

Piece	Plan	Monthly
Frontend	Vercel Hobby (personal/non-commercial — fine for judging)	$0
Backend	Fly.io shared-cpu-1x, 1 GB, always on, `iad`	$6.70
CV service	Fly.io shared-cpu-1x, 1 GB, always on, `iad`	$6.70
Database	Neon Free (0.5 GB storage, scales to zero after 5 min idle)	$0
Images	Existing AWS S3 bucket (Standard ≈ $0.023/GB-month)	< $0.10
Infrastructure total		**≈ $13.40 / month**
OpenAI usage	per redesign: plan + concept + suggestions + DIY	**Demo mode (pre-generated results)** — $0 while `AI_DEMO_MODE=true` (Evaluation Evidence §2)

Notes: Fly bills per second, so stopping the machines after judging stops the cost. Fly raised its RAM price on 2026-10-01 (1 GB machine $5.70 → $6.70); Frankfurt is $7.73. Neon's first query after idling has a cold start of up to a few seconds. Sources: Fly.io pricing, Vercel function limits, Neon pricing.

Steps (to run after approval)

Prerequisites the owner provides: Fly.io account (`fly auth login`), Vercel account linked to the GitHub org, Neon account, the existing S3 bucket name/region and an IAM key **scoped to that bucket**, an OpenAI key.

1. **Database** — `neonctl projects create --name roomify --region-id aws-us-east-1`, then `neonctl connection-string --project-id <id>` for host, database, user and password.
2. **CV service** (private) — `roomify_cv/fly.toml` is committed: no public service, `CV_HOST="::"` because Fly's private network (`roomify-cv.internal`) is IPv6-only and the image default `0.0.0.0` is IPv4.

   ```
   cd roomify_cv
   fly apps create roomify-cv
   fly deploy -a roomify-cv
   fly ips list -a roomify-cv   # must be empty: no public address
   ```

3. **Backend** — `roomify_backend/fly.toml` is committed (1 GB, always on, `/api/health` check, `VISION_SERVICE_BASE_URL=http://roomify-cv.internal:8000`, `STORAGE_TYPE=s3`).

```
cd roomify_backend
fly apps create roomify-api
fly secrets set -a roomify-api \
  SPRING_DATASOURCE_URL='jdbc:postgresql://<neon-host>/<db>?sslmode=require'
DB_USERNAME=<user> DB_PASSWORD=<password> \
  JWT_SECRET=$(openssl rand -hex 32) OPENAI_API_KEY=<key> \
  S3_BUCKET=<bucket> AWS_REGION=<region> AWS_ACCESS_KEY_ID=<id>
AWS_SECRET_ACCESS_KEY=<secret>
fly deploy -a roomify-api
curl https://roomify-api.fly.dev/api/health
```

Demo mode: `fly.toml` sets `AI_DEMO_MODE=true`, so the plan, visual concept, suggestions and DIY guides come from the bundled demo library (fitted to the user's furniture decisions and budget; the concept is a stock photo for the room type) and every result is marked `source: "demo"`. With `AI_DEMO_MODE=false` the app calls OpenAI and, because `AI_DEMO_FALLBACK` defaults to `true`, still answers from the library if a call fails (401, 429, timeout, invalid output); each fallback is logged as `AI demo result used: feature=… case=… reason=…`. Neon requires TLS, so the JDBC URL is set whole with `sslmode=require` (it overrides `DB_HOST` / `DB_PORT` / `DB_NAME`). **Long AI calls:** Fly's proxy closes a connection after 60 s with no bytes; an image edit can be silent for ~2 minutes, so `fly.toml` sets `http_options.idle_timeout = 600`. A Fly community report says this setting may not take effect: the smoke test (step 5) must include a visual concept. If it is cut off, the fix is to make the concept endpoint asynchronous (202 + poll).

4. **Frontend** — Vercel → *Add New Project* → import `Bootcamp2Roomify/roomify_frontend` → framework Next.js → environment variable `ROOMIFY_API_URL=https://roomify-api.fly.dev` → *Deploy*. The project API bridge sets `maxDuration = 300` so AI calls are not cut at Vercel's non-Fluid 10 s default.
5. **Smoke test** — run the live browser suite against the deployment: `LIVE_BASE_URL=https://<project>.vercel.app LIVE_AI=1 npx playwright test -c playwright.live.config.ts`.
6. **After judging** — `fly scale count 0 -a roomify-api -a roomify-cv` (stops billing), or `fly apps destroy`.

Risks

- `gpt-4o-mini` (default text model) is shut down by OpenAI on **2027-01-14**; switch to `gpt-6-luna` with `AI_TEMPERATURE=` and `AI_REASONING_EFFORT=low` before then (config only).
- Vercel Hobby is for non-commercial use; move to Pro ($20/month) for anything beyond judging.
- A public demo can spend the OpenAI budget: set a monthly usage limit in the OpenAI dashboard.